

2. time learning process of AI depends
 ↳ SLOPE
 ↳ measure the slope → derivative

1. check slope
 2. walk slightly forward
 3. again check slope
 4. walk slightly backward

→ ML

derivative → feel slope
 gradient descent → walks downhill
 goal → minimize error loss

A2 model → mistakes → in which dir should we move the weights so that error becomes smaller.
 optimization

Derivative → Can move 100km in 2 hours
 How fast something changes
 Speed = $\frac{\text{distance}}{\text{time}} = \frac{100}{2} = 50 \text{ km/hr}$

↳ Rate of change

M → 20°C, S → 32°C

↳ $f(n)$ divides $f'(n)$
 $f(n) \rightarrow f'(n)$
 $f'(n) = \lim_{h \rightarrow 0} \frac{f(n+h) - f(n)}{h}$

↳ $n = 3.0002$
 $n+h = 3.0002$
 tiny amount

$f(n) = n^2$
 $f'(n) = \frac{d}{dn}(n^2) = 2n$
 $S = \frac{d}{dt}, v = \frac{dx}{dt}$

→ $f(n) = n^4 \Rightarrow f'(n) = 4n^3 = 4n^3$
 → $f(n) = n^5 \Rightarrow f'(n) = 5n^4 = 5n^4$
 → $f(n) = n^1 \Rightarrow f'(n) = 1 \times n^0 = 1$

↳ Differentiation
 ↳ Integration

$f(n) = n^2 \Rightarrow n=2$
 $2 \times 2 \times 2 = 16$
 $4 \times 2 \times 2 \times 2 = 32$
 $f'(n) = 5n^4$
 $f(n) = 7, f'(n) = 0$
 constant $f'(n) = 0$
 diff. coeff. work

if $f(n) = g(n) + h(n)$
 $f'(n) = g'(n) + h'(n)$

if $f(n) = g(n) - h(n)$
 $f'(n) = g'(n) - h'(n)$

$f(n) = n^2 + n^2 + n + 1$
 $f'(n) = 2n + 2n + 1 + 0 \Rightarrow 3n^2 + 2n + 1$

$f(n) = 4n^4 + 8n^2 + 2$
 $f'(n) = 16n^3 + 16n + 0 \Rightarrow 24n^3 + 16n$

↳ ML loss like $L(w) = w^2 - 4w + 4$
 gradient of loss $\Rightarrow \frac{dL(w)}{dw} = 2w - 4$
 $w=1 \Rightarrow 2 \times 1 - 4 = -2$
 $w=2 \Rightarrow 2 \times 2 - 4 = 0$

$f(n) = 3n^4 - 5n^2 + 2n - 10$
 $f'(n) = 12n^3 - 10n + 2$

$f(n) = n^2$

90°
 180°
 270°
 360°

100°
 200°
 300°

100°
 200°
 300°

100°
 200°
 300°

100°
 200°
 300°

Product Rule: $u \cdot v$
 $f(n) = n^2 e^n$
 $f'(n) = 2n e^n + n^2 e^n$
 $(uv)' = u'v + uv'$

↳ chain rule repeatedly

Quotient rule
 $f(n) = \frac{u}{v}$
 $f'(n) = \frac{v u' - u v'}{v^2}$
 $\Rightarrow \frac{n^3 + 2(2n) - (n^2 + 1)(3n^2)}{(n^2 + 1)^2}$

Sigmoid function
 $\sigma(n) = \frac{1}{1 + e^{-n}}$ → logistic function
 $0 < \sigma(n) < 1$ → perfect for probability
 $\sigma'(n) = \frac{d}{dn} \left(\frac{1}{1 + e^{-n}} \right) = \frac{e^{-n}}{(1 + e^{-n})^2}$
 $1 - \sigma(n) = \frac{e^{-n}}{1 + e^{-n}}$

$\sigma(n)(1 - \sigma(n))$
 $\sigma'(n) = \sigma(n)(1 - \sigma(n))$
 $y = \frac{u}{v}$
 $\Rightarrow \frac{u}{v} = \frac{1}{1 + e^{-n}}$
 $\frac{(1 + e^{-n})(0) - 1(-e^{-n})}{(1 + e^{-n})^2} \Rightarrow \frac{e^{-n}}{(1 + e^{-n})^2}$
 $\sigma(n) = \frac{1}{1 + e^{-n}}$
 $1 - \sigma(n) = \frac{e^{-n}}{1 + e^{-n}}$
 $\sigma'(n) = \frac{1}{1 + e^{-n}}$
 $n=0 \Rightarrow \frac{1}{1 + e^0} = 0.5$
 $\sigma'(n) = (1 - \sigma(n)) \sigma(n)$
 $\sigma'(n) = (1 - 0.5)(0.5) = 0.25$

Chain rule: function, inside function.
 $\frac{d}{dx} f(g(x)) \Rightarrow f'(g(x)) g'(x)$
 $\frac{d}{dx} (3x+1)^5$
 $u = 3x+1$
 $u^5 \rightarrow 5u^4$
 $\frac{d}{dx} (3x+1)^5 = 5(3x+1)^4 \cdot 3 = 15(3x+1)^4$

Backpropagation: repeated chain rule.
 $\frac{dL}{dw} = \frac{dL}{da} \cdot \frac{da}{dz} \cdot \frac{dz}{dw}$
 Sends the error backward & use derivatives to improve weights.

$z = w \cdot x + b$ → bias.
 weighted sum.
 weight input
 $a = \sigma(z)$
 actual production
 $(1 - 0.7) = 0.3$

Input → Output X
 Input → Hidden → Output
 chain rule repeatedly.
 $L = f(g(h(u)))$
 $\frac{dL}{du} = \frac{dL}{df} \cdot \frac{df}{dg} \cdot \frac{dg}{dh} \cdot \frac{dh}{du}$